

**Equipment Procedure
for Determination of Furanic Compounds and Passivators in Electrical
Insulating Liquids By HPLC(Methanol Extraction Method)
High Performance Liquid Chromatography (HPLC)**

**Equipment Number: EQ-09, EQ-10, EQ-11, & EQ-12
Serial Number: 8343219, 8343856, 8344427, & 6514790**



1. PURPOSE

This standard test method procedure is intended to provide chemist / analyst / technician the general guide on performing the test analysis appropriately.

2. SCOPE

This standard operating procedure is employed in the GCC Oil Lab primarily for determination of furfural compounds in transformer oil and other related chromatographic tests which require HPLC-DAD unit system.

3. TERMS AND DEFINITIONS

3.1 High Performance Liquid Chromatography (HPLC) – a chromatographic technique that separates a mixture of compounds to identify and purify the individual component of the mixture.

4. RESPONSIBILITIES

- 4.1. Use, care, and maintenance of the High-Performance Liquid Chromatography (HPLC) Thermo Fischer Scientific is directly assigned to the end user who is usually the Sr. laboratory chemist. This responsibility may be shared to other qualified user(s) upon the consent of the Sr. Laboratory chemist.
- 4.2. Application of this responsibility is based on the Memorandum Receipt and further supported by a Memorandum of Authorization both signed by the General Manager.

5. SAFETY PRECAUTIONS

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5.1 Use a proper electrical outlet of correct voltage and current handling capacity.

5.2 Disconnect from the power supply prior to maintenance and servicing.

5.3 Ensure that the operating switch operates properly.

5.4 Lab coat, Goggles, and gloves are a must.

5.5 Working under fume hood is a must.

6. PRINCIPLES OF OPERATION

6.1 General Principle

The use of Thermo Fischer Scientific High Performance Liquid Chromatography is fully automatic with software / hardware, pump, injector, detector, column, and degasser are integrated in one operating unit system. The operator simply must familiarize with the software, prepare the test samples, place it in test chamber, select the appropriate specification or test method for the test, download the program method and then start the test sequence. Results are evaluated after each run is completed.

7. OPERATIONAL DETAILS

7.1 General Instructions

7.1.1 Turn on all instruments entire system (PC, pump, detector, auto sampler, solvent Manager, and Column compartment by pressing the on/off button).

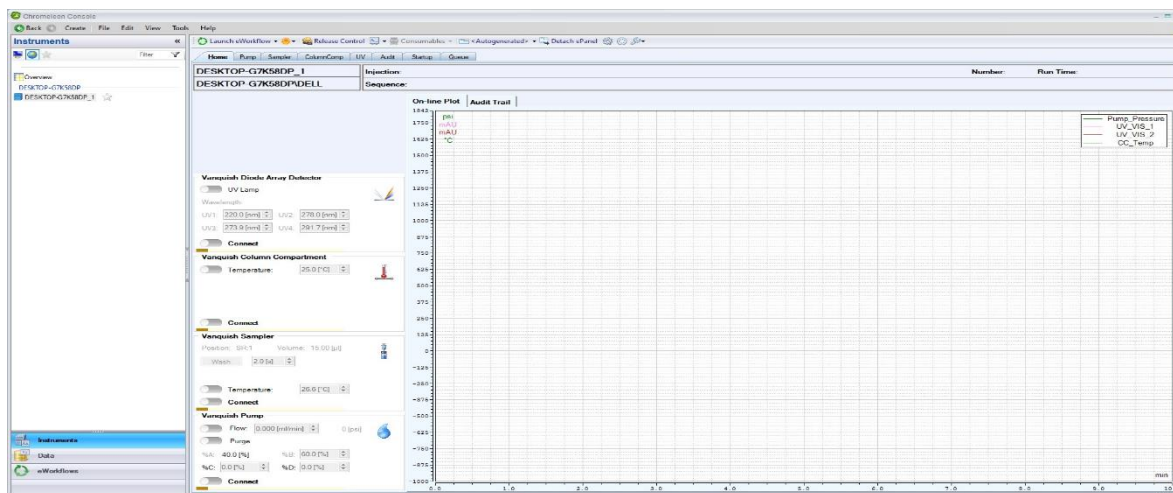
7.1.2 Click Chromeleon Manager Icon in the screen in the bottom right corner and check if the HPLC is connected to the PC.

7.1.3 Open the Chromeleon software.

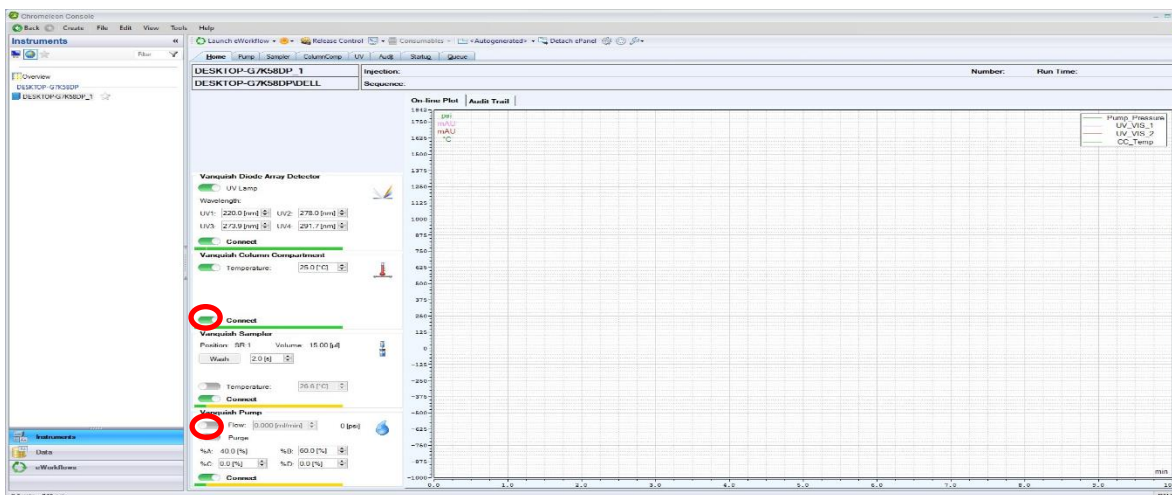
7.1.4 The Chromeleon software Screen appears with menu of commands.

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7.1.6 To connect each unit go to Instruments > Home > Connect each unit by pressing on the toggle button in each compartment box.



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7.1.7. Once the entire unit is completely connected, different commands can now be perform like setting up instrument, building method, building sequence, reprocess results and calibration of methods.

7.2 Setting Up HPLC for analysis

7.2.1 Click on Instruments and choose purge in the Pump box.

7.2.2 Set up the flow for 3ml/min and keep the gold knob open in the Pump compartment below for at least 1 min until all bubbles purged from the pump tube.

7.2.3 Stop the purge and make sure the knob closed properly.

7.2.4 Set the flow rate at 0.5ml/min.

7.2.5 Prepare the samples/Standard and place them into either rack (B, R, Y, G)

7.3 Setting Up Method Or Sequence for analysis

7.3.1 Click Data and the sequence list should appear. Each Test has separate sequence list select as appropriate.

7.3.2 Add new samples to the sequence by pressing on “Click here to add a new injection”.

7.3.3 Select the desired method from the instrument method and the processing method columns.

7.3.4 Enter the sample ID, Position, and Type (Unknown, Blank or Calibration Standard).

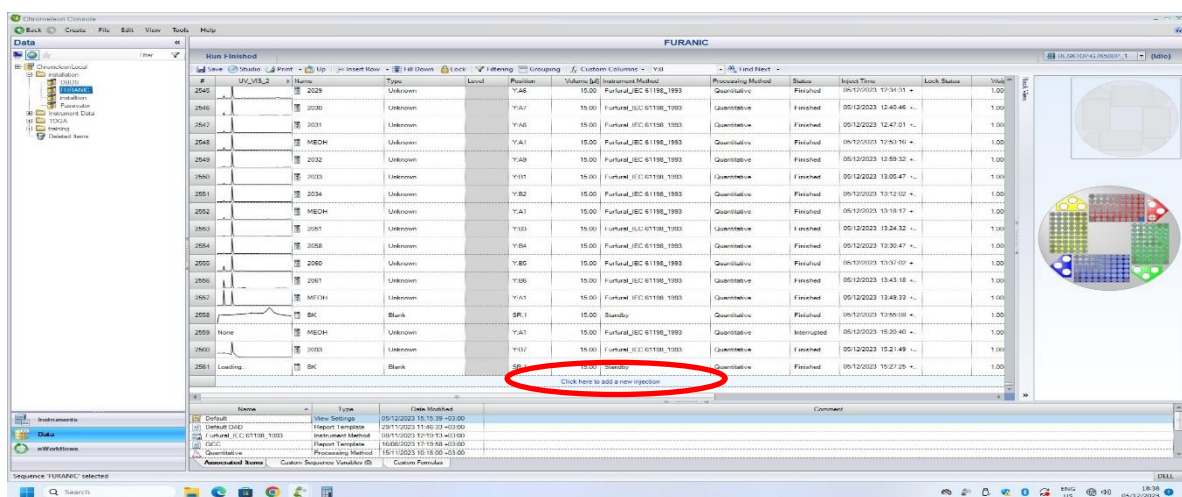
Unknown = any sample with unknown concentration.

Blank = Run with only mobile phase

Calibration Standard = Intended for new calibration curve.

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7.3.5 Save any change and Press Start/Resume. While the analysis is in progress, new samples can be added to the sequence by repeating steps 7.3.3 and 7.3.4 then save for updating the sequence.

7.3.6 After the analysis is complete, check the chromatogram by comparing the analyte with the QC sample along with their UV-Vis spectras to verify the correct compound.

7.4 Building Method for Specific Analysis

7.4.1 There is two types of methods:

Instrument method: for instrument parameters (Pump program, UV-Vis program, Autosampler program, etc..)

Processing method: for processing of the results regarding integration and calibration curve.

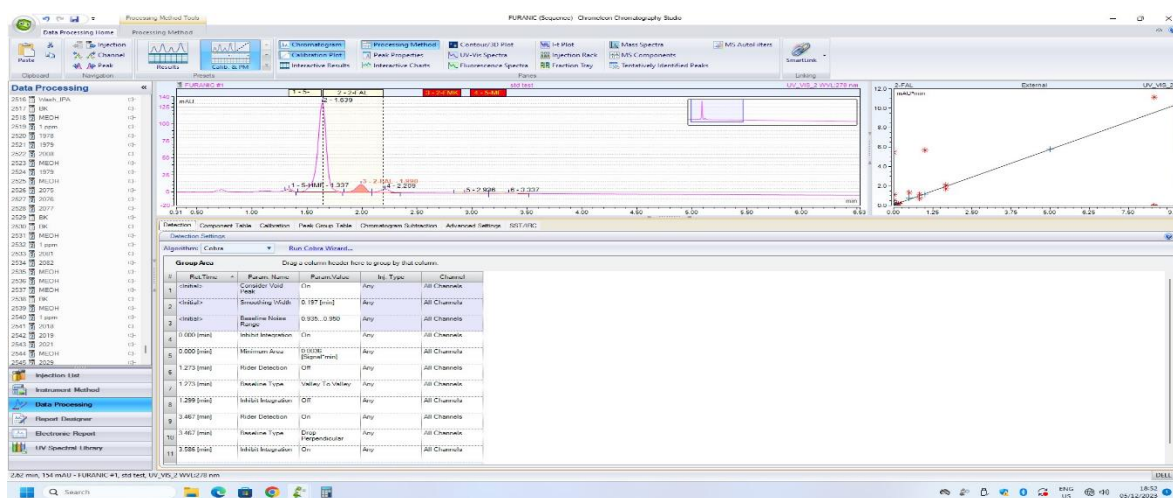
7.4.2 From the bottom of the Data page or the top ribbon select Instrument method.

7.4.3 Select create new method and click ok, instrument selection will appear, select the HPLC Instrument and click ok.

7.4.4 Entry the method name and select the data channel A, click next and instrument control parameters will window will appear

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7.4.4 Fill in all the required parameters of the window like; pump configuration, pump program, detector and instrument time events and just click next till its finish and save the method.



8.0 Standards Preparation

- 8.1 Prepare stock of 1000ppm of each compound separately.
- 8.2 Dilute to 5 different concentrations (0.05, 0.1, 0.5, 1, 5ppm) of all 5 furanic compounds mixed.
- 8.3 Into a 10-15mL test tube or Vial, take 2 mL of sample and 5 mL of methanol HPLC grade.
- 8.4 Take the tubes/vials to a vortex shaker for 2 minutes (Or normal shaker at 300rpm for 30mins)
- 8.5 Place the Tubes/Vials into a centrifuge for 5mins at 4000rpm.
- 8.6 Two Layers will form top is methanol and bottom is oil.
- 8.7 Take 1-1.5mL of the top layer into an amber vial.
- 8.8 Inject the standards in HPLC into desired method.
- 8.9 Change type to "Calibration Standard" and level for each standard.
- 8.10 Evaluate the chromatogram and calibrate the method.

9.0 Sample Preparation for Furan and Passivator

- 9.1 Into a 10-15mL test tube or Vial, take 2 mL of sample and 5 mL of methanol HPLC grade.
- 9.2 Take the tubes/vials to a vortex shaker for 2 minutes (Or normal shaker at 300rpm for 30mins)
- 9.3 Place the Tubes/Vials into a centrifuge for 5mins at 4000rpm.

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- 8.7 Two Layers will form top is methanol and bottom is oil.
- 8.8 Take 1-1.5mL of the top layer into an amber vial.
- 8.9 Prepare a sequence, run methanol twice, then QC.
- 8.10 Check QC in the QC chart then add all other samples to be analyzed in sequence.
- 8.11 After analysis is completed, integrate the peaks and record the results up to three decimals.
- 8.12 Run “wash” method then “Idle” method at the end of the sequence.

10 Calibration and Verification

- 10.1 A yearly commissioning service preventive maintenance is provided by Naizak Engineer for highly technical checkup.
- 10.2 Column Check Up
- 10.2.1 Monitor the column pressure and record during the run of same method / analysis with same parameters. Pressure should be always same or within the range.
- 10.2.2 If possible, wash the column with appropriate solvent once column performance is not good.
- 10.3 Verification is done by using standard prepared in-house or a sample of known concentration and recording in the quality control chart.
- 10.4 Column conditioning is preformed every week.

11 Method Parameters:

Method	UV-Vis	Mobile phase	Injected volume
Furfural_IEC61198_1993	UV-Vis Channel 1: 282 nm	30% Methanol	15uL
Passivator_IEC60666	UV-Vis Channel 1 (BTA): 256 nm UV-Vis Channel 2 (Irg-39): 262 nm	40% Methanol	15uL

12 REFERENCE

- 11.1 IEC 61198 (Mineral insulating oils- methods for the determination of 2- furfural and related compounds)
- 11.2 Chameleon Software Tutorial (User's Guide)

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